

NAIROBI SCHOOL 2025 PREDICTION



KCSE ANTICIPATION EXAM



231/2

BIOLOGY

PAPER 2 (THEORY)

TIME: 2 HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

INSTRUCTIONS TO CANDIDATES

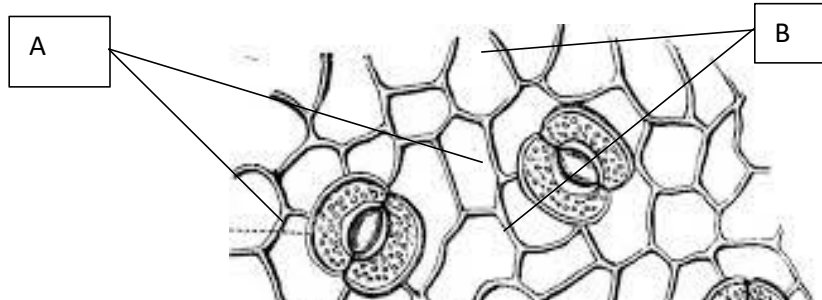
- a) This paper consists of two sections, *A* and *B*.
- b) Answer all the questions in section *A* in the spaces provided.
- c) In section *B* answer question 6(**compulsory**) and either question 7 or 8 in the spaces provided

For Examiner's use only

Section	Question	Maximum score	Candidate's score
A	1	8	
	2	8	
	3	8	
	4	8	
	5	8	
B	6	20	
	7	20	
	8	20	
Total score		80	

SECTION A (40 MARKS)

1. Study the diagram below and answer the questions that follow



(a) Explain what would happen to the cells labelled a sunny day.

(3marks)

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(b) State structural differences between cells labelled A and B.

(3marks)

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(c) Name the plant hormone that promotes the closure of the stomata.

(1mark)

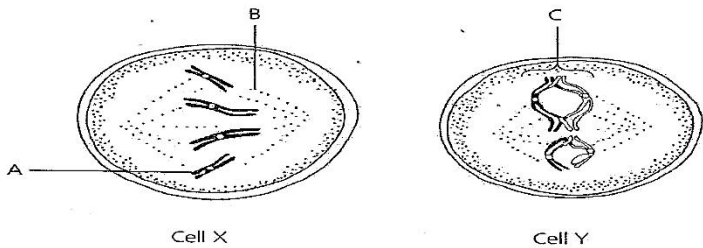
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(d) Name the site where light independent stage occurs.

(1mark)

.....

2. The diagram below show two cells, X and Y from the organism. Study the diagrams and answer the questions that follow.



(a) Name the parts labelled A, B and C.

(3 marks)

A

B

C

(b)(i) Which cell is dividing by meiosis?

(1 mark)

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(ii) Give **two** reasons for your answer in b ii above.

(2 marks)

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.....

(c) How are frogs adapted to carry out fertilisation?

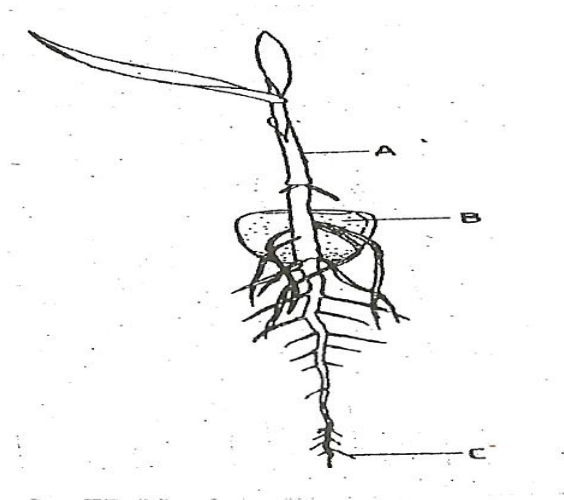
(2 marks)

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3. The diagram below represents a certain seedling.



(a) Name the parts labelled A and C.

(2 marks)

A

C

(b) State the functions of the parts labelled A, B and C.

A.

(1 mark)

.....

B.

(2 marks)

.....

.....

C.

(1mark)

.....

(c) Why is seed dormancy necessary?

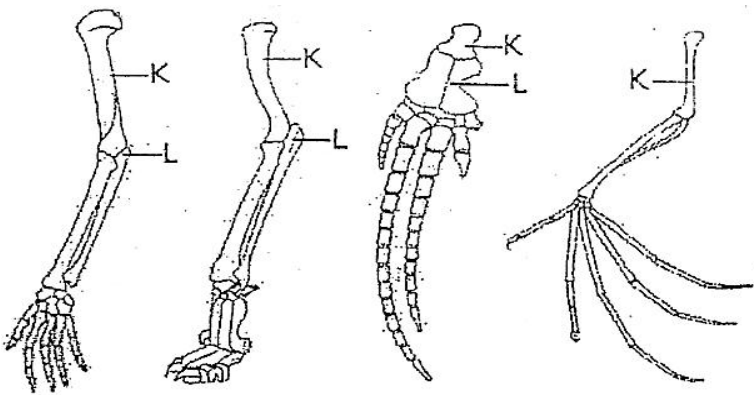
(2 marks)

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4. Study the diagrams below and answer the questions that follow.



(a) Explain the type of evolutionary evidence illustrated above.
(4 marks)

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(b) Name the parts labelled **K** and **L**.
(2 marks)

K

L

(c) Name the joint formed:
(i) At the proximal end of bone **K**.
(1 mark)

.....

(ii) At the distal end of bone **K**.
(1 mark)

.....

5. **Afia** and Kalisha were carrying out a field activity during Biology lesson. They accidentally injured themselves and they were given First Aid to stop bleeding. Afia stopped bleeding after 10 minutes while Kalisha continued bleeding regardless of further medical attention.

(a) **Name** the condition that Kalisha was likely suffering from.

(1 mark)

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(b) **Explain** the process that occurred to stop Afia's bleeding.

(3marks)

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(c) Kalisha's mother was phenotypically normal for the condition that she was suffering from. Work out a genetic cross between Kalisha's parents (4marks)

SECTION B (40 Marks)

Answer question 6 (**Compulsory**) and either question 7 or 8 in the spaces provided after question 8.

6. An investigation was carried out in a terrestrial ecosystem. The population size and species biomass were determined and recorded as follows.

Species	Population size	Species biomass(tons)
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A	1	X	10^3	1	X	10^3
B	1	X	10^3	1	X	10^{-1}
C	1	X	10^5	1	X	10
D	1	X	10	1	X	10^4

(a) What is a food chain?

(1 marks)

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(b) If all the organisms in the table had a feeding relationship, construct a food chain involving all the organisms.

(1 mark)

.....

(c) From the food chain, identify the part of the biotic component to which each of the species belongs.

(4 marks)

A.....

B

C

D

(d) State the effect on the other species if species C was removed from the ecosystem (2 marks)

A

.....

B

.....

(e) From the data construct in the species below a labelled diagram of:

(i) Pyramid of numbers;

(1 mark)

(ii) Pyramid of biomass;
(1 mark)

(f) How would you describe the pyramid of numbers?
(3 marks)

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(g) What possible feeding relationship exists for this ecosystem between the species of:

(i) The first and second trophic level;
(1 mark)

.....

(ii) The second and third trophic level.
(1 mark)

.....

(h) Explain why water logging of soil may lead to death of plants.
(3 marks)

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(b) State two protozoan diseases in man.
(2 marks)

.....

7. Describe the process of hearing in man.

(20 marks)

8. (a) Explain how high blood sugar is regulated in the body

(4marks)

(b) Describe secondary thickening in flowering plants.

(16marks)